Humana. **Claims Payment Policy**

Subject: Laboratory – Pathogen Panel Testing

Application: Medicare Advantage Products **Published date:** 10/2026

Policy number: LAB2026004

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Medicare Advantage Payment Policy

As a Medicare Advantage plan, we have developed payment policies to explain coverage more fully for services. The criteria outlined will apply to Medicare Advantage plans except when the procedure(s) has established criteria through the Original Medicare National Coverage Determinations (NCDs) and Local Coverage Determinations (LCDs) as outlined below.

In addition to the policy, claims payments are subject to other plan requirements for the processing and payment of claims, including, but not limited to, requirements of medical necessity and reasonableness and applicable referral or authorization requirements.

- In the outpatient setting, panel testing of up to five respiratory pathogens (CPT code 87428***, CPT code 87812**) **IS REIMBURSABLE** for any of the following**:**
  - For all ages:
    - Ageusia
    - Anorexia
    - Anosmia
    - Chills
    - Cough
    - Diarrhea
    - Fatigue
    - Fever
    - Headaches
    - Myalgia
    - Pharyngitis
    - Rhinorrhea
    - Vomiting
    - Weakness
  - For less than 18 years of age:
    - Apnea episodes
    - Chest retractions or nasal flaring
    - Crackles
    - Cyanosis
    - Decreased activity
    - Dehydration
    - Drooling
    - Ear tugging or otalgia
    - Fussiness in infants

- Grunting
- Irritability
- Lethargy
- Nausea
- Poor feeding in infants
- Rash
- Refusal to eat
- Stomach pain
- Tachypnea
- Vomiting after coughing
- Wheezing

- In the outpatient setting, the following molecular detection-based panel testing **IS NOT REIMBURSABLE:**
  - Testing for pathogens in the bloodstream (CPT code 0441U, CPT code 0610U)
  - Testing for pathogens of a wound (CPT code 0600U*)

Applicable procedures for Medicare Advantage plans

Procedure code	Code description
87428***	Infectious agent antigen detection by immunoassay technique, (eg, enzyme immunoassay [EIA], enzyme-linked immunosorbent assay [ELISA], fluorescence immunoassay [FIA], immunochemiluminometric assay [IMCA]) qualitative or semiquantitative; severe acute respiratory syndrome coronavirus (eg, SARS-CoV, SARS-CoV-2 [COVID-19]) and influenza virus types A and B
87812**	Infectious agent antigen detection by immunoassay with direct optical (ie, visual) observation; severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) (coronavirus disease [COVID-19]) and influenza virus types A and B
0441U	Infectious disease (bacterial, fungal, or viral infection), semiquantitative biomechanical assessment (via deformability cytometry), whole blood, with algorithmic analysis and result reported as an index Proprietary test: IntelliSep® test Lab/ Manufacturer: Cytovale®
0600U*	Infectious disease (wound infection), identification of 65 organisms and 30 antibiotic resistance genes, wound swab, real-time PCR, reported as positive or negative for each organism Proprietary test: FidaLab Molecular Wound Infection Test Lab/Manufacturer: FidaLab LLC
0610U	Infectious disease (antimicrobial susceptibility), phenotypic antimicrobial susceptibility testing of positive blood culture using microfluidic sensor technology to quantify bacterial growth response to multiple antibiotic types, reporting categorical susceptibility (susceptible, susceptible dose dependent, intermediate, resistant), minimum inhibitory concentration, and interpretive comments Proprietary test: LifeScale Gram Negative Kit (LSGN) with the LifeScale AST system Lab/Manufacturer: Affinity Biosensors, LLC

** The payment criteria for 0600U apply to all Medicare jurisdictions, except states awarded to CGS Administrators, LLC, Noridian Healthcare Solutions, LLC, Palmetto GBA, and Wisconsin Physicians Service Insurance Corporation.*
*** The payment criteria for 87812 apply to all Medicare jurisdictions, except states awarded to First Coast Service Options, Inc. and Novitas Solutions, Inc.*
****The payment criteria for 87428 apply to all Medicare jurisdictions, except states awarded to First Coast Service Options, National Government Services, Inc. and Novitas Solutions, Inc.*

Original Medicare Local Coverage Determinations (LCDs)

Contractor name	Policy title	Policy number	Procedure code
CGS Administrators, LLC	MolDX: Molecular Diagnostic Tests (MDT)	L36021	0600U
Palmetto GBA	MolDX: Molecular Diagnostic Tests (MDT)	L35025	0600U
Wisconsin Physicians Service Insurance Corporation	MolDX: Molecular Diagnostic Tests (MDT)	L36807	0600U
Noridian Healthcare Solutions, LLC	MolDX: Molecular Diagnostic Tests (MDT)	L35160	0600U
National Government Services, Inc	Respiratory Pathogen Panel Testing	L39027	87428
Novitas Solutions, Inc.	Respiratory Pathogen Panel Testing	L38916	87428 87812
First Coast Service Options, Inc.	Respiratory Pathogen Panel Testing	L38918	87428 87812

References

AAP. Respiratory Syncytial Virus. *Red Book: 2021–2024 Report of the Committee on Infectious Diseases*. 2021:0. doi:10.1542/9781610025782-S3_114

ACG. ACG updates clinical guidelines on prevention, diagnosis, treatment of C. difficile infection. ACP Gastroenterology Monthly Newsletter: American College of Gastroenterology; 2021

Almonacid DE, Kraal L, Ossandon FJ, et al. 16S rRNA gene sequencing and healthy reference ranges for 28 clinically relevant microbial taxa from the human gut microbiome. *PLOS ONE*. 2017;12(5):e0176555. doi:10.1371/journal.pone.0176555

American Society for Microbiology. MolDX: Multiplex Nucleic Acid Amplified Tests for RespiratoryViral Panels (DL37301). https://www.amp.org/AMP/assets/File/position-statements/2017/JointCommentLettertoNoridioanJEforMultiplexViralPanelTests-Respiratory-DL37301.pdf

Armstrong D, Meyr A. Principles of acute wound management. Updated May 29, 2025. https://www.uptodate.com/contents/principles-of-acute-wound-management

ASCP. ASCP Expands List of Commonly Used Tests Physicians and Patients Should Question. https://www.ascp.org/news/news-details/2019/09/10/ascp-expands-list-of-commonly-used-tests-physicians-and-patients-should-question

ASM. Making Sense of Respiratory Viral Panel Results. Updated March 2, 2020. https://asm.org/articles/2020/march/making-sense-of-respiratory-viral-panel-results

Baddour LM, Chen AF. Periprosthetic joint infection: Epidemiology, microbiology, clinical manifestations, and diagnosis. Updated March 23, 2026. https://www.uptodate.com/contents/periprosthetic-joint-infection-epidemiology-microbiology-clinical-manifestations-and-diagnosis

Baghdadi JD, Tung CC, Johnson JK, Morgan DJ, Harris AD. Changes in use of multiplex respiratory panel testing during the COVID-19 pandemic. *Infect Control Hosp Epidemiol*. Oct 10 2024:1-6. doi:10.1017/ice.2024.163

Banerjee R, Teng CB, Cunningham SA, et al. Randomized Trial of Rapid Multiplex Polymerase Chain Reaction-Based Blood Culture Identification and Susceptibility Testing. *Clin Infect Dis*. Oct 1 2015;61(7):1071-80. doi:10.1093/cid/civ447

Beal SG, Tremblay EE, Toffel S, Velez L, Rand KH. A Gastrointestinal PCR Panel Improves Clinical Management and Lowers Health Care Costs. *J Clin Microbiol*. Jan 2018;56(1)doi:10.1128/jcm.01457-17

BioCode. FDA-Cleared Gastrointestinal Pathogen Panel (GPP). Updated 2026. https://www.apbiocode.com/gi_panel.htm

BioCode. FDA-Cleared Respiratory Pathogen Panel (RPP). Updated 2026. https://apbiocode.com/rpp_panel.htm

BioFire. The BioFire® FilmArray® Blood Culture Identification (BCID) Panel. Updated 2026. https://www.biofiredx.com/products/the-filmarray-panels/filmarraybcid/

BioFire. The BioFire® FilmArray® Gastrointestinal (GI) Panel. Updated 2026. https://www.biofiredx.com/products/the-filmarray-panels/filmarraygi/

BioFire. The BioFire® FilmArray® Meningitis/Encephalitis (ME) Panel. Updated 2026. https://www.biofiredx.com/products/the-filmarray-panels/filmarrayme/
BioFire. The BioFire® FilmArray® Respiratory 2.1 (RP2.1) Panel. Updated 2026. https://www.biofiredx.com/products/the-filmarray-panels/filmarrayrp/
bioMérieux. The BioFire® Joint Infection (JI) Panel. Updated 2026. https://www.biomerieux.com/us/en/our-offer/clinical-products/biofire-joint-infection-panel.html
Bonkat G, Bartoletti R, Bruyere F, et al. European Association of Urology (EAU) Guidelines on Urological Infections. Updated March 2026. http://uroweb.org/guideline/urological-infections/#3
Bonnin P, Miszczak F, Kin N, et al. Study and interest of cellular load in respiratory samples for the optimization of molecular virological diagnosis in clinical practice. journal article. August 09 2016;16(1):384. doi:10.1186/s12879-016-1730-9
Brendish NJ, Poole S, Naidu VV, et al. Clinical impact of molecular point-of-care testing for suspected COVID-19 in hospital (COV-19POC): a prospective, interventional, non-randomised, controlled study. *The Lancet Respiratory Medicine*. 2020;8(12):1192-1200. doi:10.1016/S2213-2600(20)30454-9
Buss SN, Leber A, Chapin K, et al. Multicenter evaluation of the BioFire FilmArray gastrointestinal panel for etiologic diagnosis of infectious gastroenteritis. *J Clin Microbiol*. Mar 2015;53(3):915-25. doi:10.1128/jcm.02674-14
Caliendo AM. Multiplex PCR and Emerging Technologies for the Detection of Respiratory Pathogens. *Clinical Infectious Diseases*. 2011;52(suppl_4):S326-S330. doi:10.1093/cid/cir047
CDC. About Bacterial Meningitis. Updated September 9, 2025. https://www.cdc.gov/meningitis/about/bacterial-meningitis.html
CDC. About Fungal Meningitis. Updated September 9, 2025. https://www.cdc.gov/meningitis/about/fungal-meningitis.html
CDC. About Meningococcal Disease. Updated February 1, 2024. https://www.cdc.gov/meningococcal/about/index.html
CDC. About Respiratory Illnesses. Updated August 18, 2025. https://www.cdc.gov/respiratory-viruses/about/index.html
CDC. About Rhinoviruses. Updated February 19, 2026. https://www.cdc.gov/rhinoviruses/about/index.html
CDC. About Sepsis. Updated August 19, 2025. https://www.cdc.gov/sepsis/about/
CDC. About Strep Throat. Updated January 15, 2026. https://www.cdc.gov/group-a-strep/about/strep-throat.html
CDC. Clinical Guidance for Patients with Acute Respiratory Illness Being Hospitalized When SARS-CoV-2 and Influenza Viruses are Co-Circulating. Updated December 6, 2023. https://www.cdc.gov/flu/hcp/clinical-guidance/testing-guidance-for-clinicians-hospitalized.html
CDC. Clinical Testing and Diagnosis for CDI. Updated March 6, 2024. https://www.cdc.gov/c-diff/hcp/diagnosis-testing/index.html
CDC. Multiplex Assays Authorized for Simultaneous Detection of Influenza Viruses and SARS-CoV-2 by FDA. CDC. Updated December 19, 2025. https://www.cdc.gov/flu/hcp/testing-methods/flu-covid19-detection.html
CDC. Respiratory Viruses and Young Children. Updated August 18, 2025. https://www.cdc.gov/respiratory-viruses/risk-factors/young-children.html
CDC. RSV in Infants and Young Children. Updated February 20, 2026. https://www.cdc.gov/rsv/infants-young-children/index.html
Chang S-S, Hsieh W-H, Liu T-S, et al. Multiplex PCR System for Rapid Detection of Pathogens in Patients with Presumed Sepsis – A Systemic Review and Meta-Analysis. *PLOS ONE*. 2013;8(5):e62323. doi:10.1371/journal.pone.0062323
Claas EC, Burnham CA, Mazzulli T, Templeton K, Topin F. Performance of the xTAG(R) gastrointestinal pathogen panel, a multiplex molecular assay for simultaneous detection of bacterial, viral, and parasitic causes of infectious gastroenteritis. *Journal of microbiology and biotechnology*. 2013;23(7):1041-5
CMS. Local Coverage Determination (LCD): Foodborne Gastrointestinal Panels Identified by Multiplex Nucleic Acid Amplification Tests (NAATs) (L37766). https://www.cms.gov/medicare-coverage-database/details/lcd-details.aspx?LCDId=37766
Couturier MR, Lee B, Zelyas N, Chui L. Shiga-toxigenic Escherichia coli detection in stool samples screened for viral gastroenteritis in Alberta, Canada. *J Clin Microbiol*. Feb 2011;49(2):574-8. doi:10.1128/jcm.01693-10
Cybulski RJ, Jr., Bateman AC, Bourassa L, et al. Clinical Impact of a Multiplex Gastrointestinal Polymerase Chain Reaction Panel in Patients With Acute Gastroenteritis. *Clin Infect Dis*. Nov 13 2018;67(11):1688-1696. doi:10.1093/cid/ciy357
Dando SJ, Mackay-Sim A, Norton R, et al. Pathogens penetrating the central nervous system: infection pathways and the cellular and molecular mechanisms of invasion. *Clin Microbiol Rev*. Oct 2014;27(4):691-726. doi:10.1128/cmr.00118-13
Diagnostics R. cobas® eplex Blood Culture Identification Panels. Updated 2026. https://diagnostics.roche.com/us/en/products/params/eplex-bcid-panels.html
Diasorin. NxTAG® Respiratory Pathogen Panel v2. Updated 2026. https://us.diasorin.com/en/molecular-diagnostics/kits-reagents/nxtag-respiratory-pathogen-panel-V2

Esteban J, Salar-Vidal L, Schmitt BH, et al. Multicenter evaluation of the BIOFIRE Joint Infection Panel for the detection of bacteria, yeast, and AMR genes in synovial fluid samples. *J Clin Microbiol*. Nov 21 2023;61(11):e0035723. doi:10.1128/jcm.00357-23

Fernandez-Soto P, Sanchez-Hernandez A, Gandasegui J, et al. Strong-LAMP: A LAMP Assay for Strongyloides spp. Detection in Stool and Urine Samples. Towards the Diagnosis of Human Strongyloidiasis Starting from a Rodent Model. *PLoS Negl Trop Dis*. Jul 2016;10(7):e0004836. doi:10.1371/journal.pntd.0004836

GenetWorx. Wounds Pathogen Panel. Updated 2026. https://www.genetworx.com/services/wound-pathogen-panel

GenMark. Respiratory Pathogen (RP) Panel and NEW Respiratory Pathogen Panel 2 (RP2). Updated 2026. https://www.medical-xprt.com/products/genmark-respiratory-pathogen-panels-750159

Ginocchio CC. Detection of respiratory viruses using non-molecular based methods. *Journal of clinical virology : the official publication of the Pan American Society for Clinical Virology*. Nov 2007;40 Suppl 1:S11-4. doi:10.1016/s1386-6532(07)70004-5

Goldenberg DL, Sexton DJ. Septic arthritis of native joints in adults. Updated August 1, 2025. https://www.uptodate.com/contents/septic-arthritis-of-native-joints-in-adults

Gupta K. Acute complicated urinary tract infection (including pyelonephritis) in adults. Updated November 25, 2025. https://www.uptodate.com/contents/acute-complicated-urinary-tract-infection-including-pyelonephritis-in-adults

Gyawali B, Ramakrishna K, Dhamoon AS. Sepsis: The evolution in definition, pathophysiology, and management. *SAGE Open Med*. 2019;7:2050312119835043. doi:10.1177/2050312119835043

Hansen LS, Lykkegaard J, Thomsen JL, Hansen MP. Acute lower respiratory tract infections: Symptoms, findings and management in Danish general practice. *Eur J Gen Pract*. Dec 2020;26(1):14-20. doi:10.1080/13814788.2019.1674279

Hanson KE, Azar MM, Banerjee R, et al. Molecular Testing for Acute Respiratory Tract Infections: Clinical and Diagnostic Recommendations From the IDSA's Diagnostics Committee. *Clin Infect Dis*. Dec 17 2020;71(10):2744-2751. doi:10.1093/cid/ciaa508

Hayden M, Hanson K, Englund J, et al. Infectious Diseases Society of America Guidelines on the Diagnosis of COVID-19: Molecular Diagnostic Testing. Infectious Diseases Society of America. Updated September 6, 2023. https://www.idsociety.org/practice-guideline/covid-19-guideline-diagnostics/

Healthcare O. QuickFinder™ COVID-19 / Flu Antigen Self Test. Updated 2026. https://www.osanghc.com/en/product/detail?idx=116

Hill AT, Gold PM, El Solh AA, Metlay JP, Ireland B, Irwin RS. Adult Outpatients With Acute Cough Due to Suspected Pneumonia or Influenza: CHEST Guideline and Expert Panel Report. *Chest*. Jan 2019;155(1):155-167. doi:10.1016/j.chest.2018.09.016

Humphrey JM, Ranbhise S, Ibrahim E, et al. Multiplex Polymerase Chain Reaction for Detection of Gastrointestinal Pathogens in Migrant Workers in Qatar. 2016;95(6):1330-1337. doi:10.4269/ajtmh.16-0464

Humphries RM, Linscott AJ. Laboratory diagnosis of bacterial gastroenteritis. *Clin Microbiol Rev*. Jan 2015;28(1):3-31. doi:10.1128/cmr.00073-14

Kelly CR, Fischer M, Allegretti JR, et al. ACG Clinical Guidelines: Prevention, Diagnosis, and Treatment of Clostridioides difficile Infections. *Official journal of the American College of Gastroenterology | ACG*. 2021;116(6):1124-1147. doi:10.14309/ajg.0000000000001278

La Hoz RM, Morris MI. Intestinal parasites including Cryptosporidium, Cyclospora, Giardia, and Microsporidia, Entamoeba histolytica, Strongyloides, Schistosomiasis, and Echinococcus: Guidelines from the American Society of Transplantation Infectious Diseases Community of Practice. *Clin Transplant*. Sep 2019;33(9):e13618. doi:10.1111/ctr.13618

Lamy B, Sundqvist M, Idelevich EA. Bloodstream infections - Standard and progress in pathogen diagnostics. *Clin Microbiol Infect*. Feb 2020;26(2):142-150. doi:10.1016/j.cmi.2019.11.017

Leber W, Lammel O, Siebenhofer A, Redlberger-Fritz M, Panovska-Griffiths J, Czypionka T. Comparing the diagnostic accuracy of point-of-care lateral flow antigen testing for SARS-CoV-2 with RT-PCR in primary care (REAP-2). *eClinicalMedicine*. 2021;38doi:10.1016/j.eclinm.2021.101011

Lewinski MA, Alby K, Babady NE, et al. Exploring the Utility of Multiplex Infectious Disease Panel Testing for Diagnosis of Infection in Different Body Sites: A Joint Report of the Association for Molecular Pathology, American Society for Microbiology, Infectious Diseases Society of America, and Pan American Society for Clinical Virology. *The Journal of Molecular Diagnostics*. 2023;25(12):857-875. doi:10.1016/j.jmoldx.2023.08.005

Liesenfeld O, Lehman L, Hunfeld KP, Kost G. Molecular diagnosis of sepsis: New aspects and recent developments. *European journal of microbiology & immunology*. 2014;4(1):1-25. doi:10.1556/EuJMI.4.2014.1.1

Liu J, Kabir F, Manneh J, et al. Development and assessment of molecular diagnostic tests for 15 enteropathogens causing childhood diarrhoea: a multicentre study. *The Lancet Infectious diseases*. Aug 2014;14(8):716-724. doi:10.1016/s1473-3099(14)70808-4

Luminex. xTAG® Gastrointestinal Pathogen Panel (GPP). Updated 2026. https://www.luminexcorp.com/gastrointestinal-pathogen-panel/
McCarty TP, Cumagun P, Meeder J, et al. Test Performance and Potential Clinical Utility of the GenMark Dx ePlex Blood Culture Identification Gram-Negative Panel. *Microbiol Spectr*. Feb 14 2023;11(1):e0409222. doi:10.1128/spectrum.04092-22
McDonald LC, Gerding DN, Johnson S, et al. Clinical Practice Guidelines for Clostridium difficile Infection in Adults and Children: 2017 Update by the Infectious Diseases Society of America (IDSA) and Society for Healthcare Epidemiology of America (SHEA). *Clin Infect Dis*. Mar 19 2018;66(7):e1-e48. doi:10.1093/cid/cix1085
Medical Diagnostics. One Vial, Multiple Pathogens. Updated 2026. https://www.mdlab.com/
Meyrier A. Sampling and evaluation of voided urine in the diagnosis of urinary tract infection in adults. Updated Febraury 28, 2025. https://www.uptodate.com/contents/sampling-and-evaluation-of-voided-urine-in-the-diagnosis-of-urinary-tract-infection-in-adults
MicroGenDX. Wound Care Updated March 2, 2026. https://microgendx.com/wound-care/
Miller JM, Binnicker MJ, Campbell S, et al. Guide to Utilization of the Microbiology Laboratory for Diagnosis of Infectious Diseases: 2024 Update by the Infectious Diseases Society of America (IDSA) and the American Society for Microbiology (ASM). *Clin Infect Dis*. Mar 5 2024;doi:10.1093/cid/ciae104
MircoGenDX. UroKEY. Updated 2026. https://patients.microgendx.com/products/urokey
Mormeneo Bayo S, López González E, Bellés Bellés A, et al. Detection and pathological role of intestinal protozoa in children. *Parasitol Int*. Jun 2022;88:102558. doi:10.1016/j.parint.2022.102558
O'Grady NP, Alexander E, Alhazzani W, et al. Society of Critical Care Medicine and the Infectious Diseases Society of America Guidelines for Evaluating New Fever in Adult Patients in the ICU. *Critical care medicine*. 2023;51(11):1570-1586. doi:10.1097/ccm.0000000000006022
Onori M, Coltella L, Mancinelli L, et al. Evaluation of a multiplex PCR assay for simultaneous detection of bacterial and viral enteropathogens in stool samples of paediatric patients. *Diagnostic microbiology and infectious disease*. Jun 2014;79(2):149-54. doi:10.1016/j.diagmicrobio.2014.02.004
Operario DJ, Houpt E. Defining the causes of diarrhea: novel approaches. *Current opinion in infectious diseases*. Oct 2011;24(5):464-71. doi:10.1097/QCO.0b013e32834aa13a
Osmon DR, Berbari EF, Berendt AR, et al. Diagnosis and Management of Prosthetic Joint Infection: Clinical Practice Guidelines by the Infectious Diseases Society of Americaa. *Clinical Infectious Diseases*. 2012;56(1):e1-e25. doi:10.1093/cid/cis803
Palavecino E. One Sample, Multiple Results The Use of Multiplex PCR for Diagnosis of Infectious Syndromes. Webpage. *Clinical Laboratory News*. 2019;2018(11/1)doi:10.1016/j.idc.2019.10.006
Pammi M. Neonatal bacterial sepsis: Clinical features and diagnosis in neonates born at less than 35 weeks gestation. Updated March 2, 2026. https://www.uptodate.com/contents/neonatal-bacterial-sepsis-clinical-features-and-diagnosis-in-neonates-born-at-less-than-35-weeks-gestation
Pathnostics. Guidance UTI. Updated 2026. https://pathnostics.com/tests/guidance-uti/#1647110828433-7f31d2bd-54d7
Petti CA, Polage CR. Molecular diagnosis of central nervous system infections. Updated July 30, 2025. https://www.uptodate.com/contents/molecular-diagnosis-of-central-nervous-system-infections
QIAGEN. QIAstat-Dx Respiratory SARS-CoV-2 Panel. Updated 2026. https://www.qiagen.com/us/products/diagnostics-and-clinical-research/infectious-disease/qiastat-dx-syndromic-testing/qiastat-dx-eua-us/
QuidelOrtho. QuickVue® COVID-19 Test. Updated 2026. https://www.quidelortho.com/global/en/consumers/quickvue-otc
Ravn C, Neyt J, Benito N, et al. Guideline for management of septic arthritis in native joints (SANJO). *J Bone Joint Infect*. 2023;8(1):29-37. doi:10.5194/jbji-8-29-2023
Rhodes A, Evans LE, Alhazzani W, et al. Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock: 2016. *Critical care medicine*. Mar 2017;45(3):486-552. doi:10.1097/ccm.0000000000002255
Riddle MS, DuPont HL, Connor BA. ACG Clinical Guideline: Diagnosis, Treatment, and Prevention of Acute Diarrheal Infections in Adults. *The American journal of gastroenterology*. May 2016;111(5):602-22. doi:10.1038/ajg.2016.126
Rubin DT, Ananthakrishnan AN, Siegel CA, Barnes EL, Long MD. ACG Clinical Guideline Update: Ulcerative Colitis in Adults. *The American journal of gastroenterology*. Jun 3 2025;120(6):1187-1224. doi:10.14309/ajg.0000000000003463
Scallan E, Griffin PM, Angulo FJ, Tauxe RV, Hoekstra RM. Foodborne illness acquired in the United States--unspecified agents. *Emerging infectious diseases*. Jan 2011;17(1):16-22. doi:10.3201/eid1701.091101p2
Scallan Walter EJ CZ, Tierney R, Griffin PM, Hoekstra RM, Payne DC, et al. Foodborne Illness Acquired in the United States—Major Pathogens, 2019. *Emerging Infectious Diseases*. 2025;31(4):669-677doi:10.3201/eid3104.240913

Schultz G, Bjarnsholt T, James GA, et al. Consensus guidelines for the identification and treatment of biofilms in chronic nonhealing wounds. *Wound Repair Regen*. Sep 2017;25(5):744-757. doi:10.1111/wrr.12590

Shane AL, Mody RK, Crump JA, et al. 2017 Infectious Diseases Society of America Clinical Practice Guidelines for the Diagnosis and Management of Infectious Diarrhea. *Clin Infect Dis*. Nov 29 2017;65(12):1963-1973. doi:10.1093/cid/cix959

Surawicz CM, Brandt LJ, Binion DG, et al. Guidelines for diagnosis, treatment, and prevention of Clostridium difficile infections. *The American journal of gastroenterology*. Apr 2013;108(4):478-98; quiz 499. doi:10.1038/ajg.2013.4

Thomas M, Bomar PA. Upper Respiratory Tract Infection. *StatPearls*. StatPearls Publishing LLC.; 2026. https://pubmed.ncbi.nlm.nih.gov/30422556/

Uyeki TM, Bernstein HH, Bradley JS, et al. Clinical Practice Guidelines by the Infectious Diseases Society of America: 2018 Update on Diagnosis, Treatment, Chemoprophylaxis, and Institutional Outbreak Management of Seasonal Influenzaa. 2018;doi:10.1093/cid/ciy866

V. Wintzingerode F, Göbel UB, Stackebrandt E. Determination of microbial diversity in environmental samples: pitfalls of PCR-based rRNA analysis. 1997;21(3):213-229. doi:doi:10.1111/j.1574-6976.1997.tb00351.x

van Asten SAV, Boers SA, de Groot JDF, Schuurman R, Claas ECJ. Evaluation of the Genmark ePlex® and QIAstat-Dx® respiratory pathogen panels in detecting bacterial targets in lower respiratory tract specimens. *BMC Microbiol*. Aug 26 2021;21(1):236. doi:10.1186/s12866-021-02289-w

Veritor B. BD Veritor™ Plus System for Rapid Detection of SARS-CoV-2 & Flu A+B. Updated 2026. https://bdveritor.bd.com/en-us/products/cov2-flu-a-b

Visseaux B, Le Hingrat Q, Collin G, et al. Evaluation of the QIAstat-Dx Respiratory SARS-CoV-2 Panel, the First Rapid Multiplex PCR Commercial Assay for SARS-CoV-2 Detection. *Journal of Clinical Microbiology*. 2020;58(8):e00630-20. doi:10.1128/JCM.00630-20

Ward C, Stocker K, Begum J, Wade P, Ebrahimsa U, Goldenberg SD. Performance evaluation of the Verigene(R) (Nanosphere) and FilmArray(R) (BioFire(R)) molecular assays for identification of causative organisms in bacterial bloodstream infections. *European journal of clinical microbiology & infectious diseases : official publication of the European Society of Clinical Microbiology*. Mar 2015;34(3):487-96. doi:10.1007/s10096-014-2252-2

Watts GS, Youens-Clark K, Slepian MJ, et al. 16S rRNA gene sequencing on a benchtop sequencer: accuracy for identification of clinically important bacteria. *Journal of applied microbiology*. 2017;123(6):1584-1596. doi:10.1111/jam.13590

Weiss SL, Peters MJ, Alhazzani W, et al. Surviving Sepsis Campaign International Guidelines for the Management of Septic Shock and Sepsis-Associated Organ Dysfunction in Children. *Pediatr Crit Care Med*. Feb 2020;21(2):e52-e106. doi:10.1097/pcc.0000000000002198

WHO. Diarrhoeal disease. Updated March 7, 2024. https://www.who.int/news-room/fact-sheets/detail/diarrhoeal-disease

Woods CR, Bradley JS, Chatterjee A, et al. Clinical Practice Guideline by the Pediatric Infectious Diseases Society (PIDS) and the Infectious Diseases Society of America (IDSA): 2023 Guideline on Diagnosis and Management of Acute Bacterial Arthritis in Pediatrics. *Journal of the Pediatric Infectious Diseases Society*. 2023;13(1):1-59. doi:10.1093/jpids/piad089

Wu KAK, D.N.; Seidelman, J.L.; Seyler, T.M. Native Joint Septic Arthritis. *Antibiotics* 2024;13, 596

Xie J, Kim K, Berenger BM, et al. Comparison of a Rapid Multiplex Gastrointestinal Panel with Standard Laboratory Testing in the Management of Children with Hematochezia in a Pediatric Emergency Department: Randomized Controlled Trial. *Microbiol Spectr*. Jun 15 2023;11(3):e0026823. doi:10.1128/spectrum.00268-23

Yan Y, Zhang S, Tang YW. Molecular assays for the detection and characterization of respiratory viruses. *Seminars in respiratory and critical care medicine*. Aug 2011;32(4):512-26. doi:10.1055/s-0031-1283288

Zhan Z, Guo J, Xiao Y, et al. Comparison of BioFire FilmArray gastrointestinal panel versus Luminex xTAG Gastrointestinal Pathogen Panel (xTAG GPP) for diarrheal pathogen detection in China. *Int J Infect Dis*. Oct 2020;99:414-420. doi:10.1016/j.ijid.2020.08.020

Zhang H, Morrison S, Tang YW. Multiplex polymerase chain reaction tests for detection of pathogens associated with gastroenteritis. *Clinics in laboratory medicine*. Jun 2015;35(2):461-86. doi:10.1016/j.cll.2015.02.006